General Physics

Unit 2: Vectors and Projectile Motion

Student Workbook

Name: _____ Period: _____

How to use this workbook

- There is one section for each lesson. Each section lists the lesson's learning targets, the key ideas, and practice problems. Show all work in the space below each problem.
- Every learning target sits in one of four tiers: **Foundations**, **Core Skills**, **Target**, and **Extension**. A higher tier counts only when the tiers below it are secure, and you can reassess any tier after targeted practice.
- Problems marked **Check yourself** are a quick test of the section's main idea. If you cannot do one, revisit the key ideas before moving on.
- Use the target tracker at the back after every section and before every quiz.
- Each section starts with a **worked example**. Read it first. The next problem is **partly worked**: fill in the blanks, then try the rest on your own.
- Course conventions: kinematic equations start from positions, such as $x_f = \frac{1}{2}at^2 + v_i t + x_i$ (we set $x_i = 0$; $y_i = 0$ for a launch from the ground, and $y_i = h$ with $y_f = 0$ for a launch from a height h). Displacement is $\Delta x = x_f - x_i$; when a question asks for it, the answer is written as Δx or Δy. Down is negative, and $a_y = -9.8\ \mathrm{m/s^2}$. Ignore air resistance unless a problem says otherwise.

Every problem in a section or quiz prep page has a mark next to its number:

●	**Core**	Everyone does these. Together they cover the section's learning targets.
○	**More practice**	Extra reps. Do them when a core problem gave you trouble, or before a quiz.
★	**Challenge**	Harder, multi-step problems for when the core problems feel easy.

Contents

Section	Title	Tier
2.1	Scalars and Vectors	Foundations
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2.5	Reference Frames and Relative Velocity	Core Skills
	Get Ready for Quiz 1	Quiz 1: Vectors and Relative Velocity
2.6	Horizontal Launches: x and y Columns	Core Skills
2.7	Horizontal Launch Practice	Core Skills
2.8	Velocity During Flight	Core Skills
2.9	Marble off the Table Lab	Core Skills
	Get Ready for Quiz 2	Quiz 2: Horizontal Launches
2.10	Angled Launches: Split and Symmetry	Target
2.11	Level-Ground Launches with Desmos	Target
	Unit 2 Review	All tiers
	Projectile Launch Project: Planning Pages	Target
	Reference sheet, vocabulary, target tracker, extension menu	

Reference sheet

Vectors

Vector components	Resultant vectors
$A_x = A\cos\theta$ $A_y = A\sin\theta$ θ is measured from the +x axis. Left and down components are negative.	Add components: $R_x = A_x + B_x, R_y = A_y + B_y$ Magnitude: $R = \sqrt{R_x^2 + R_y^2}$ Direction: $\theta = \tan^{-1}\left(\frac{\lvert R_y\rvert}{\lvert R_x\rvert}\right)$

SOH-CAH-TOA: $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}, \cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}, \tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

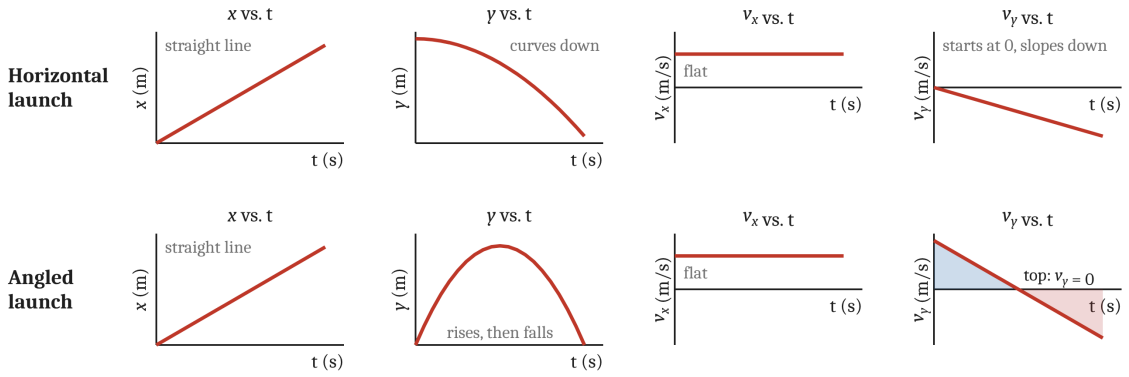
Pythagorean theorem: $c^2 = a^2 + b^2$

Relative velocity

- Add velocities as vectors: $v_{\text{object rel. ground}} = v_{\text{object rel. medium}} + v_{\text{medium rel. ground}}$
- **Assume:** each velocity is constant, and the medium (water or air) carries the object along with it.
- **Perpendicular velocities:** find the resultant with the Pythagorean theorem and its direction with $\tan^{-1}$.
- **Crossing time:** only the velocity component straight across the river sets it.
- **Drift:** the current carries the boat downstream while it crosses.
- **Extension, landing directly across:** aim upstream so the upstream part of the boat's velocity cancels the current.

Motion graphs

Top row: a horizontal launch. Bottom row: an angled launch on level ground. In the angled v_y graph, the shaded area above the axis is the rise and the area below is the fall.



Projectile motion: x world and y world

Ignore air resistance. Down is negative. Time t is the same in both columns and links them.

x world (horizontal)	y world (vertical)
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x = 0$	$a_y = -9.8\ \text{m/s}^2$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =
Every projectile	
Assume: $a_x = 0$, so $v_{fx} = v_{ix}$ the whole flight. **Equation:** $x_f = v_{ix}t + x_i$	**Assume:** $a_y = -9.8\ \text{m/s}^2$. At the top, $v_y = 0$. **Equations:** $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ $v_{fy} = a_y t + v_{iy}$ $v_{fy}^2 = v_{iy}^2 + 2a_y(y_f - y_i)$
Horizontal launch	
Assume: $v_{ix} =$ launch speed, $x_i = 0$.	**Assume:** $v_{iy} = 0$, $y_i = h$ (launch height), $y_f = 0$ (the ground).
Angled launch from the ground	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$. Range uses the total time.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = 0$, $y_f = 0$. It lands with $v_{fy} = -v_{iy}$.
Extension: angled launch from a cliff	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = h$ (cliff height), $y_f = 0$ (the ground). v_{fy} is negative at landing.

Solving routine

1. Sketch the path. Label the launch point, the landing point, and the ground ($y = 0$).
2. Fill in both columns. Mark what you know and what you want.
3. Find t from the column that has enough knowns (usually the y world).
4. Use that t in the other column to finish.

Common mistakes

- $v_{iy} = 0$ only for a horizontal launch.
- a_x is always 0. Only the y world speeds up or slows down.
- Cosine goes with the side next to the angle: $v_{ix} = v_i \cos\theta$ when θ is measured from the horizontal.
- At the top, $v_y = 0$ but v_x is not, so the ball is still moving.
- v_f is not 0 at the ground. The equations describe the ball just before it lands, while it is still falling fast; the stop happens during the impact, which the equations do not model. Use $y_f = 0$, not $v_{fy} = 0$.

Velocity at any moment

$v_x = v_{ix}$ the whole flight and $v_y = a_y t + v_{iy}$, so $v = \sqrt{v_x^2 + v_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|v_y|}{v_x}\right)$ (below the horizontal when v_y is negative, above it when positive).

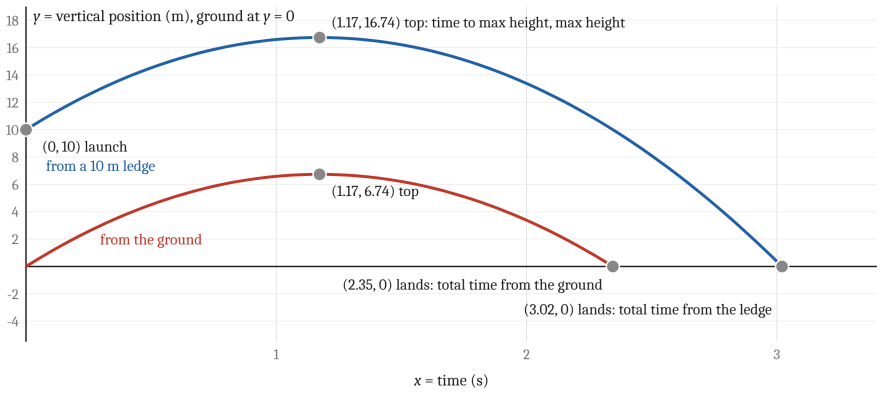
Solving for time with Desmos

When you know the heights but not the time, the y-world equation $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ has t^2 in it. Desmos solves it by graphing: you click where the curve reaches the final height. Example: a ball launched at 15 m/s at $50°$ ($v_{iy} = 11.49$ m/s) from the ground, and, as an Extension, from the top of a 10 m ledge.

Method 1: graph it in Desmos

Desmos (desmos.com/calculator) graphs the y-world equation so you can click to read off key points. In Desmos, x stands for time t and y stands for vertical position.

1. Set Desmos to degrees: click the wrench (top right) and choose **Degrees**.
2. Type the y-world equation with your numbers, using x for time (the pattern is $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$). From the ground: `y=0.5(-9.8)x^2+15sin(50)x+0`. From a height, the last number is the launch height: `y=0.5(-9.8)x^2+15sin(50)x+10`.
3. Zoom out until you see the whole arc. Click the curve: gray points appear. Click a point to see its coordinates (time, height).
4. With the ground at $y = 0$, the ball lands where the curve crosses the x axis. To land on a ledge instead, add the ledge height as a second line, such as `y=6`, and click where it crosses the curve.
5. For the horizontal distance, type v_{ix} times the time you found: `15cos(50)*2.345` gives 22.6 m from the ground, and `15cos(50)*3.021` gives 29.1 m from the ledge.



To find	Click on	Example reading
Time to max height and max height	the top of the curve	Ground: (1.17, 6.74). Ledge: (1.17, 16.74), so 16.74 m above the ground.
Total time, launched from the ground	where the curve crosses the x axis again	(2.35, 0): 2.35 s
Total time, launched from a height	where the curve crosses the x axis (the ground)	(3.02, 0): 3.02 s
Range or horizontal final position	type v_{ix} × time on a new line	Ground: 22.6 m. Ledge: 29.1 m.

- Horizontal launch: $v_{iy} = 0$, so type `y=0.5(-9.8)x^2+h` with the launch height for h, and click where the curve crosses `y=0`.
- Time to the top another way: graph v_y with `y=-9.8x+15sin(50)`. It crosses the x axis at the time of max height.

Vocabulary

Term	Definition
Scalar	A quantity with magnitude only (distance, speed, time)
Vector	A quantity with magnitude and direction (change in position, velocity, acceleration)
Magnitude	The size of a vector, always positive
Direction	The way a vector points, given as an angle from a reference axis
Resultant	The single vector equal to the sum of two or more vectors
Component	The horizontal (x) or vertical (y) part of a vector
Angle of launch	The angle of the initial velocity above or below the horizontal
Projectile	An object moving through the air under gravity alone
Trajectory	The curved path a projectile follows
Range	The horizontal change in position from launch to landing
Independence of motion	Horizontal and vertical motions happen at the same time but do not affect each other
Relative velocity	An object's velocity as measured from a particular reference frame, such as the riverbank or the moving water

Section 2.1 · Scalars and Vectors

Tier: Foundations **In class:** Vocabulary sort; scalar and vector practice

Learning targets

- Key terms I can tell a scalar from a vector and use magnitude, direction, and resultant correctly.

Key ideas

- A **scalar** has magnitude only (distance, speed, time, mass).
- A **vector** has magnitude and direction (change in position, velocity, acceleration, force).
- Change in position (displacement) starts from positions: $\Delta x = x_f - x_i$. It can be negative; distance traveled cannot.
- Signs show direction: right and up are positive, left and down are negative.

Worked example

A student walks 30 m east, then 10 m west. The walk takes 20 s. Take east as positive. Find the distance traveled, the change in position, the average speed, and the average velocity.

1. Distance adds every step, ignoring direction: $30\ \text{m} + 10\ \text{m} = 40\ \text{m}$.
2. Change in position compares start and end: $\Delta x = +30\ \text{m} + (-10\ \text{m}) = +20\ \text{m}$.
3. Average speed = distance ÷ time = $40\ \text{m} \div 20\ \text{s} = 2.0\ \text{m/s}$.
4. Average velocity = Δx ÷ time = $+20\ \text{m} \div 20\ \text{s} = +1.0\ \text{m/s}$.

Answer: 40 m; $\Delta x = +20$ m (20 m east); 2.0 m/s; +1.0 m/s (1.0 m/s east)

1. ● Label each quantity S (scalar) or V (vector).

a) 25 m/s ___ b) 25 m/s north ___ c) 3.0 s ___ d) 9.8 m/s^2 down ___

e) 12 km ___ f) 12 km southwest ___ g) mass, 4.0 kg ___ h) temperature, 21 °C ___

2. ● **Partly worked.** A runner completes exactly one lap of a 400 m track and finishes at the starting line.

Distance traveled = _____ m.

The runner ends at the start, so $x_f = x_i$ and $\Delta x =$ _____ m.

In one sentence, explain why the two answers are different.

3. ● A car drives 60 km east, then 20 km west. The whole trip takes 1.0 h. Take east as positive.

a) Distance traveled b) Change in position c) Average speed d) Average velocity

4. ○ A bus drives 5 km north to a stop, then 5 km south back to the station. Is its distance traveled zero? Is its change in position zero? Explain.

5. ● An elevator goes up 12 m, then down 20 m. Take up as positive. Find the distance traveled and the change in position Δy.

6. ○ Which of these needs a direction to be useful? Explain each in a few words.

a) The weather report says the wind is 30 km/h. b) A pilot's flight plan. c) A recipe's baking time.

7. ○ A ball is thrown straight up, rises 5.0 m, and is caught at the same height it was thrown from. Find the distance it traveled and its change in position Δy.

8. ○ A swimmer swims four lengths of a 25 m pool and finishes at the starting wall. Find the distance traveled and the change in position.

9. ● Take east as positive. Write each as a signed number: a) 8 m west b) 3.0 m/s east c) 12 m west d) 0.5 m/s west

10. ○ A student says, "The car's velocity is 60 km/h." What is missing? Rewrite the statement so it describes a velocity.

11. ★ **Challenge.** Two vectors have magnitudes 6 m and 8 m. What is the largest their resultant could be? The smallest? Sketch how each one happens.

12. ● **Check yourself.** Write one sentence explaining why speed is a scalar but velocity is a vector.

Section 2.2 · Drawing and Adding Vectors

Tier: Foundations **In class:** Vector Walk lab

Learning targets

- F1 I can draw vectors to scale and add them tip-to-tail.
- F3 I can use the Pythagorean theorem and inverse tangent to find a resultant's magnitude and direction.
- F6 I can find the resultant of two perpendicular vectors.

Key ideas

- Draw each vector to scale with a ruler and protractor. Put the tail of the next vector at the tip of the last.
- The **resultant** goes from the tail of the first vector to the tip of the last.
- For perpendicular vectors: $R = \sqrt{A^2 + B^2}$ and $\theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$.
- Give direction as an angle from a reference direction, such as 37° N of E (start facing east, turn 37° toward north).

Worked example

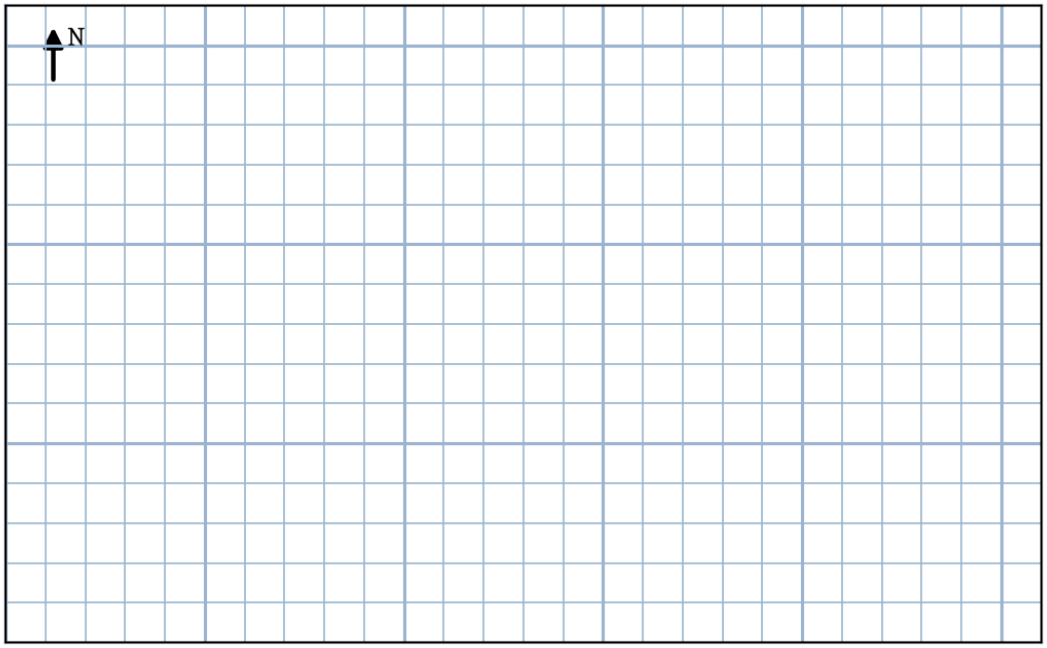
A student walks 120 m west, then 50 m south. Find the change in position (magnitude and direction).

1. Sketch: a 120 m arrow pointing left, then a 50 m arrow pointing down from its tip.
2. The legs are perpendicular, so $R = \sqrt{120^2 + 50^2} = \sqrt{16\,900} = 130$ m.
3. $\theta = \tan^{-1}(50 \div 120) = 22.6°$, measured from west toward south.
4. Check: R is longer than either leg and shorter than their sum (170 m).

Answer: 130 m at 22.6° S of W

1. ● **Scale drawing.** Use the grid (each square is 5 m). A student walks 40 m east, then 30 m north. Draw both vectors tip-to-tail, draw the resultant, and measure its length and angle.

Measured: R = _____ at _____ Calculated: R = _____ at _____



2. ● **Partly worked.** You walk 3 blocks east and then 4 blocks north.

$R = \sqrt{(___^2 + ___^2)}$ = _____ blocks

$\theta = \tan^{-1}(___ \div ___)$ = _____° N of E

3. ● A hiker walks 12 km north and then 5 km west. Find the resultant (magnitude and direction).

4. ● **Vector walk preview.** On the walk, a group goes 25 m south and then 60 m east. Find the resultant, then describe the single straight walk that would bring them back to the start.

5. ○ A plane flies 300 km east and then 400 km south. Find its change in position (magnitude and direction).

6. ○ A dog runs 8.0 m west and then 6.0 m south. Find the magnitude and direction of its resultant.

7. ○ Does the order you add vectors in change the resultant? Test it by redrawing problem 2 with north first.

8. ★ **Challenge.** You walk 5.0 m north and then some distance east. You end up 13 m from where you started. How far east did you walk?

9. ● **Check yourself.** Vectors of 5 m east and 3 m west add to what resultant?

Section 2.3 · Components: Breaking Vectors Apart

Tier: Foundations **In class:** Component practice with protractors

Learning targets

- F2 I can use SOH-CAH-TOA to resolve a vector into x and y components.
- F3 I can find a vector's magnitude and direction from its components.

Key ideas

- For a vector of magnitude A at angle θ above the +x axis: $A_x = A\cos\theta$ and $A_y = A\sin\theta$.
- Signs come from the picture: left and down are negative.
- Going back: $A = \sqrt{A_x^2 + A_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|A_y|}{|A_x|}\right)$. Then name the direction in words.
- Always sketch the triangle first. The side next to the angle uses cosine. Check that your calculator is in degrees.

Worked example

A ball is kicked at 25 m/s at 37° above the horizontal, to the right. Find the horizontal and vertical components of its velocity.

1. Sketch: an arrow up and to the right, 37° above the +x axis.
2. $v_x = 25\cos 37^\circ = 20.0$ m/s. It points right, so it is positive.
3. $v_y = 25\sin 37^\circ = 15.0$ m/s. It points up, so it is positive.
4. Check: $\sqrt{20.0^2 + 15.0^2} = 25$ m/s.

Answer: $v_x = +20.0$ m/s, $v_y = +15.0$ m/s

1. ● **Partly worked.** A ball is thrown at 20 m/s at 30° above the horizontal.

$v_{ix} = 20\cos$ ___° = _____ m/s

$v_{iy} = 20\sin$ ___° = _____ m/s

2. ● Resolve each vector into x and y components. Sketch each one.

a) 50 m/s at 37° above the horizontal

b) 15 m at 60° south of east

3. ● A soccer ball is kicked at 25 m/s at 53° above the ground. Find the horizontal and vertical components of its initial velocity, v_{ix} and v_{iy}.

4. ○ A ramp launches a marble at 3.0 m/s, 20° above the horizontal. Find v_{ix} and v_{iy}.

5. ● A ball's velocity has components $v_x = 8.0\ \text{m/s}$ and $v_y = -6.0\ \text{m/s}$. Find its speed and direction.

6. ○ A vector has components $x = -12\ \text{m}$ and $y = +5\ \text{m}$. Find its magnitude and direction.

7. ● As the launch angle of a 20 m/s ball increases from 0° to 90°, what happens to v_{ix}? To v_{iy}? At what angle are they equal?

8. ○ A plane climbs at 250 km/h at 12° above the horizontal. How fast is it moving horizontally? How fast is it gaining height?

9. ○ A hiker's change in position is 40 m at 60° south of west. Find its x and y components (east and north are positive).

10. ○ Without calculating, which component is bigger for a vector at 20° above the horizontal, x or y? Explain.

11. ★ **Challenge.** A ball leaves the ground at 40° above the horizontal with a vertical component of 12 m/s. Find its launch speed and its horizontal component.

12. ● **Check yourself.** A student writes $v_{ix} = 30 \sin 40^\circ$. When is that correct?

Section 2.4 · Resultants from Components

Tier: Core Skills **In class:** Compass-trip practice stations

Learning targets

- C1 I can add a trip of north, south, east, and west legs by adding the x parts and the y parts.
- F3 I can find a resultant's magnitude and direction from its components.

Key ideas

- Write each leg as an x part (east +, west −) or a y part (north +, south −).
- Add every x part to get R_x and every y part to get R_y.
- Then $R = \sqrt{R_x^2 + R_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|R_y|}{|R_x|}\right)$. Sketch R_x and R_y to name the direction.

Worked example

A delivery drone flies 3 km east, 4 km north, then 2 km west. How far is it from base, and in which direction?

1. Sort the legs. x parts: +3 km and −2 km. y parts: +4 km.
2. Add: $R_x = +3 + (-2) = +1$ km. $R_y = +4$ km.
3. $R = \sqrt{1^2 + 4^2} = 4.12$ km.
4. $\theta = \tan^{-1}(4 \div 1) = 76.0°$. R_x is east and R_y is north, so the drone is 76.0° N of E from base.

Answer: 4.12 km at 76.0° N of E (to fly home: 4.12 km at 76.0° S of W)

1. ● **Partly worked.** A robot drives 6 m east, 2 m north, 2 m east, then 4 m north. Fill in the table, then find the robot's change in position.

$$R = \sqrt{R_x^2 + R_y^2} = _____ \quad \theta = \tan^{-1}(|R_y| \div |R_x|) = _____$$

Leg	x part (east +)	y part (north +)
6 m east	+6 m	0
2 m north		
2 m east		
4 m north		
Sum	$R_x =$	$R_y =$

2. ● A ship sails 40 km north, 25 km east, 10 km south, then 15 km east. Find its change in position.

3. ○ A delivery robot drives 12 m north, 5 m east, 4 m south, then 1 m east. Find its change in position.

4. ● A hiker walks 2.0 km west, 3.5 km south, then 1.0 km west. How far, and in which direction, must the hiker walk to get straight back to the start?

5. ○ In a mall you walk 40 m south, 30 m west, then 10 m north. Find your change in position.

6. ○ A taxi drives 6 blocks north, 3 blocks east, 2 blocks south, 5 blocks east, then 1 block north. How far is it from where it started, and in which direction?

7. ○ A robot vacuum moves 2.5 m west, 4.0 m north, 1.5 m east, then 1.0 m south. Find its change in position.

8. ● Ana walks 30 m east, then 40 m north. Ben walks 40 m north, then 30 m east. Who ends up farther from the start? Explain without calculating.

9. ★ **Challenge.** $A = 30$ m east and $B = 20$ m at 60° N of E. B is not along a compass direction, so break it into x and y parts first (Section 2.3). Then find $A + B$.

10. ● **Check yourself.** Could 3 m east and 4 m north add to 7 m? Explain.

Section 2.5 · Reference Frames and Relative Velocity

Tier: Core Skills **In class:** Boat-and-river problems

Learning targets

- C5 I can solve relative-velocity problems with perpendicular vectors (a boat crossing a river).

Key ideas

- Velocity depends on the reference frame: the boat's velocity relative to the water is not its velocity relative to the bank.
- $v_{\text{boat rel. ground}} = v_{\text{boat rel. water}} + v_{\text{water rel. ground}}$
- Crossing a river: $t = \dfrac{\text{width}}{v_{\text{across}}}$. The current does not change that time.
- Downstream drift: $\text{drift} = v_{\text{current}}\, t$

Worked example

A boat moves at 4.0 m/s relative to the water, aimed straight across an 80 m river. The current flows at 3.0 m/s. Find the time to cross, how far downstream the boat lands, and its velocity relative to the ground.

1. Time uses only the across motion: $t = 80\ \text{m} \div 4.0\ \text{m/s} = 20\ \text{s}$.
2. Drift uses only the current: $3.0\ \text{m/s} \times 20\ \text{s} = 60\ \text{m}$ downstream.
3. The two velocities are perpendicular: $v = \sqrt{4.0^2 + 3.0^2} = 5.0\ \text{m/s}$.
4. Direction: $\theta = \tan^{-1}(3.0 \div 4.0) = 36.9°$ downstream from straight across.

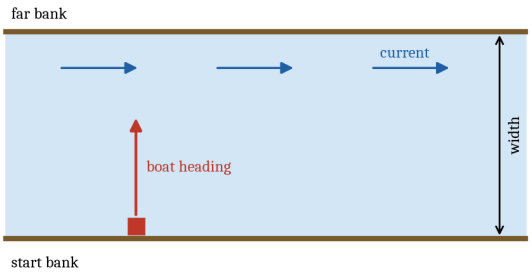
Answer: 20 s; 60 m downstream; 5.0 m/s at 36.9° downstream from straight across

1. ● **Partly worked.** A boat heads straight across a 120 m wide river at 4.0 m/s relative to the water. The current flows at 1.5 m/s.

a) t = width $\div v_{\text{across}} = 120 \div$ _____ = _____ s

b) drift = $1.5 \times$ _____ = _____ m downstream

c) $v = \sqrt{4.0^2 + 1.5^2} =$ _____ m/s at $\tan^{-1}(1.5 \div 4.0) =$ _____° downstream from straight across



2. ● A moving walkway moves at 1.0 m/s. You walk at 1.5 m/s relative to the walkway. Find your speed relative to the ground when you walk a) with the walkway and b) against it.

3. ● A swimmer swims straight across a 60 m river at 1.2 m/s relative to the water. The current is 0.50 m/s. Find the time to cross, the downstream drift, and the swimmer's speed relative to the bank.

4. ● If the current in problem 1 doubled, would the crossing time change? Would the landing spot change? Explain.

5. ● A plane points due north with an airspeed of 220 m/s. A 40 m/s wind blows toward the east. Find the plane's velocity relative to the ground.

6. ○ A train moves east at 25 m/s. A passenger walks toward the back of the train at 1.5 m/s. Find the passenger's velocity relative to the ground.

7. ○ A plane with an airspeed of 200 m/s flies into a 30 m/s headwind, then turns around and flies with it as a tailwind. Find its ground speed each way.

8. ○ A kayaker paddles straight across a 50 m river at 2.5 m/s relative to the water and lands 30 m downstream. Find the crossing time and the speed of the current.

9. ★ **Challenge.** A ferry points straight across a 300 m river. It reaches the far bank in 100 s, landing 120 m downstream. Find the ferry's speed relative to the water and the speed of the current.

10. ● **Check yourself.** To cross a river in the least time, which way should you point the boat?

Get Ready for Quiz 1

Tier: Quiz 1: Vectors and Relative Velocity **In class:** Quiz 1: Vectors and relative velocity (full period)

About this page

- Quiz 1 covers F1, F2, F3, F6, C1 (compass trips), and C5. Complete this page the night before the quiz.

Key ideas

- Rate yourself on each target in the tracker at the back of this workbook before you start.
- If a problem below gives you trouble, go back to the matching section and redo its worked example and one problem.

1. ● Define resultant and component in your own words.

2. ● Find the resultant of 9.0 m east and 12 m south (magnitude and direction).

3. ● Resolve 35 m/s at 25° below the horizontal (pointing right) into components.

4. ● A velocity has components $v_x = -5.0\ \text{m/s}$ and $v_y = +12\ \text{m/s}$. Find its magnitude and direction.

5. ● A drone flies 8.0 km east, 6.0 km north, then 2.0 km west. Find its change in position.

6. ● A canoe heads straight across a 45 m river at 3.0 m/s relative to the water; the current is 1.2 m/s. Find the time to cross, the drift, and the canoe's velocity relative to the bank.

7. ● **Scale drawing.** Using $1\text{ cm} = 10\text{ m}$, draw 50 m east followed by 30 m north. Measure the resultant, then check it with the Pythagorean theorem.

8. ○ A student walks 20 m north, 15 m east, then 5 m south. Find the change in position.

9. ○ A boat aims straight across a 100 m river at 5.0 m/s relative to the water; the current is 2.0 m/s. Find the crossing time, the drift, and the boat's speed relative to the bank.

10. ○ Resolve 18 m/s at 40° above the horizontal into components.

11. ○ A swimmer aims straight across a 40 m river at 1.0 m/s relative to the water; the current is 0.60 m/s. Find the crossing time and the drift.

12. ● Explain in one or two sentences why a faster current does not change the time to cross a river when the boat aims straight across.

13. ○ Is change in position a scalar or a vector? Give an example where distance traveled and change in position are different.

Section 2.6 · Horizontal Launches: x and y Columns

Tier: Core Skills **In class:** Horizontal launches in x and y columns; Desmos for time

Learning targets

- F4 I can explain why horizontal and vertical motions are independent.
- F5 I can state that v_x stays constant and $a_y = -9.8\ \text{m/s}^2$ during flight.
- C2 I can organize a projectile problem into x and y columns linked by the shared time.
- C3 I can find the time to land and the landing position of a horizontally launched projectile.

Key ideas

- The x and y motions are independent. The only thing they share is the time t.
- x column: v_{ix} = launch speed, $a_x = 0$, $x_i = 0$. y column: $v_{iy} = 0$, $a_y = -9.8\ \text{m/s}^2$, $y_i = h$ (launch height), $y_f = 0$ (the ground).
- Find t from the y column: $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ In Desmos, graph $y = -4.9x^2 + h$ and click where it crosses the x-axis.
- Carry t to the x column: $x_f = v_{ix}t + x_i$

Worked example

A ball rolls off a 0.80 m high table at 2.0 m/s. How long is it in the air, and how far from the table does it land?

1. x column: $v_{ix} = 2.0\ \text{m/s}$, $a_x = 0$, $x_i = 0$, $x_f = ?$. y column: $v_{iy} = 0$, $a_y = -9.8\ \text{m/s}^2$, $y_i = 0.80\ \text{m}$, $y_f = 0$, $t = ?$
2. The y column has three knowns, so it gives the time: $0 = -4.9t^2 + 0 + 0.80$.
3. Desmos: graph $y = -4.9x^2 + 0.80$. It crosses the x-axis at $x = 0.404$, so $t = 0.404$ s. (By hand: $t^2 = 0.80 \div 4.9$, $t = 0.404$ s.)
4. x column: $x_f = (2.0)(0.404) + 0 = 0.808$ m.

Answer: $t = 0.404$ s; the ball lands 0.808 m from the table

1. ● Partly worked. A ball rolls off a 1.25 m high table at 3.0 m/s. Fill in the columns, then finish.

a) y column: $0 = -4.9t^2$ + _____. Desmos crossing: t = _____ s

b) x column: $x_f = (3.0)$(_____) + 0 = _____ m

x (horizontal)	y (vertical)
$v_{ix} =$	$v_{iy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =

2. ● A stone is thrown horizontally at 15 m/s from the top of a 45 m cliff. Find the time in the air and how far from the base of the cliff it lands.

3. ○ A ball rolls off a 1.8 m shelf at 3.0 m/s. Find the time in the air and the landing distance.

4. ○ A skateboarder rolls off a 2.0 m loading dock at 5.0 m/s. How far from the dock does the skateboarder land?

5. ● Ball A is dropped from a table at the same moment ball B rolls off the same table at 2.0 m/s. Which ball hits the floor first? Explain using the y column.

6. ● Fill in the blanks for any horizontal launch: $v_{iy} =$ _____, $a_x =$ _____, $a_y =$ _____, $y_f =$ _____ (when it lands on the ground).

7. ● A plane flying horizontally at 80 m/s, 500 m above the ground, drops a supply crate. How long does the crate fall, and how far ahead of the target must the plane release it?

8. ○ A marble rolls off a 0.60 m bench at 1.2 m/s. Find the time in the air and the landing distance.

9. ○ A ball rolls off a table and lands 0.45 s later. Use the y column to find the height of the table.

10. ★ **Challenge.** A ball rolls off a 1.25 m table and lands 2.0 m from the table's edge. Find its launch speed.

11. ● **Check yourself.** Which column do you use to find the time for a horizontal launch, and why?

Section 2.7 · Horizontal Launch Practice

Tier: Core Skills **In class:** Horizontal launch practice stations; rifle TIPERs

Learning targets

- C3 I can find the time to land, the landing position, or the launch speed of a horizontally launched projectile.

Key ideas

- The height alone sets the time: a faster launch lands farther away, not sooner.
- Working backward: if you know where it lands, find t from the y column, then $v_{ix} = \dfrac{x_f - x_i}{t}$
- If you know the time from the x column, the y column gives the height: $\Delta y = \frac{1}{2} a_y t^2$ for $v_{iy} = 0$.

Worked example: finding the launch speed

A ball leaves a 1.0 m high table horizontally and lands 1.5 m from the table's edge. How fast was it rolling?

1. y column: $v_{iy} = 0, a_y = -9.8\ \text{m/s}^2, y_i = 1.0\ \text{m}, y_f = 0, t = ?$
2. Desmos: graph $y = -4.9x^2 + 1.0$. It crosses the x-axis at $t = 0.452$ s.
3. x column: $x_i = 0, x_f = 1.5\ \text{m}, t = 0.452\ \text{s}, v_{ix} = ?$
4. $v_{ix} = x_f \div t = 1.5 \div 0.452 = 3.32\ \text{m/s}$.

Answer: The ball left the table at 3.32 m/s

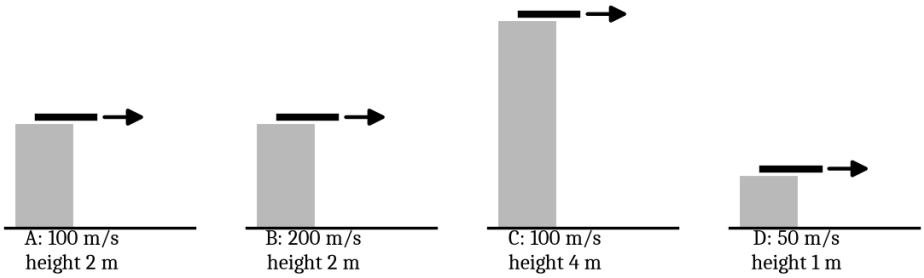
1. ● Partly worked. A ball thrown horizontally at 12 m/s from the roof of a building lands 30 m from the base. How tall is the building?

a) x column: $t = x_f \div v_{ix} = 30 \div 12 =$ _____ s

b) y column: $\Delta y = \frac{1}{2}(-9.8)($_____$)^2$ = _____ m, so the building is _____ m tall

2. ● Rifle shots (TIPERs style). Each rifle fires a bullet horizontally. Rank the times for the bullets to hit the ground, greatest first. Explain.

Ranking: _____



3. ● A ball rolls off a 0.95 m table and lands 1.1 m from the edge. Find its launch speed.

4. ● A soccer ball is kicked horizontally at 20 m/s off the edge of a 30 m cliff. Find the time in the air and where it lands.

5. ● If you double the launch speed of a ball rolling off a table, what happens to the time in the air? To the landing distance?

6. ○ A marble rolling at 2.4 m/s leaves a table and lands 1.0 m from the edge. How long was it in the air, and how high is the table?

7. ○ A diver runs off a 3.2 m platform at 2.5 m/s. How far from the edge does the diver enter the water?

8. ○ A ball rolled off a 1.0 m table lands 1.5 m out (the worked example). Where would it land if it rolled off at the same speed from a 1.0 m table on the Moon, where $a_y = -1.6\ \text{m/s}^2$? Use Desmos: graph $y = -0.8x^2 + 1.0$.

9. ○ A rock is kicked horizontally off a ledge and hits the ground 1.2 s later, 6.0 m from the base. Find its launch speed and the height of the ledge.

10. ★ **Challenge.** A marble rolls off a 1.0 m table and must land in a cup that is 0.15 m tall, with the cup's center 1.2 m from the table. How fast must the marble roll? (Hint: it only falls to the top of the cup.)

11. ● **Check yourself.** Two balls roll off the same table, one at 1.0 m/s and one at 3.0 m/s. Which lands first? Which lands farther away?

Section 2.8 · Velocity During Flight

Tier: Core Skills **In class:** Velocity components and motion graphs

Learning targets

- C4 I can find the horizontal and vertical velocity components of a projectile at any time.

Key ideas

- v_x never changes: $v_x = v_{ix}$ because $a_x = 0$.
- v_y grows downward by 9.8 m/s every second: $v_{fy} = a_y t + v_{iy}$
- Graphs: x–t is a straight line, y–t curves down, v_x–t is flat, and v_y–t is a straight line sloping down at −9.8 m/s².
- The area under a velocity–time graph is the change in position. The slope is the acceleration.

Worked example

A ball rolls off a table at 3.0 m/s. Find v_x and v_y 0.30 s after it leaves the table.

1. x column: $a_x = 0$, so v_x stays 3.0 m/s the whole flight.
2. y column: $v_{iy} = 0, a_y = -9.8\ \text{m/s}^2, t = 0.30\ \text{s}$.
3. $v_{fy} = a_y t + v_{iy} = (-9.8)(0.30) + 0 = -2.94\ \text{m/s}$ (the minus sign means downward).

Answer: $v_x = 3.0\ \text{m/s}, v_y = -2.94\ \text{m/s}$

1. ● **Partly worked.** A ball rolls off a table at 2.0 m/s. Fill in the table.

Time (s)	0	0.10	0.20	0.30	0.40
v_x (m/s)	2.0				
v_y (m/s)	0	−0.98			

2. ● A stone is thrown horizontally at 8.0 m/s from a cliff. Find v_x and v_y after 1.5 s.

3. ○ A ball leaves a table at 1.8 m/s and is in the air for 0.45 s. Find v_x and v_y just before it lands.

4. ● Two balls roll off the same table, one fast and one slow. Just before landing, how do their v_y values compare? Which one is moving at a steeper angle? Explain.

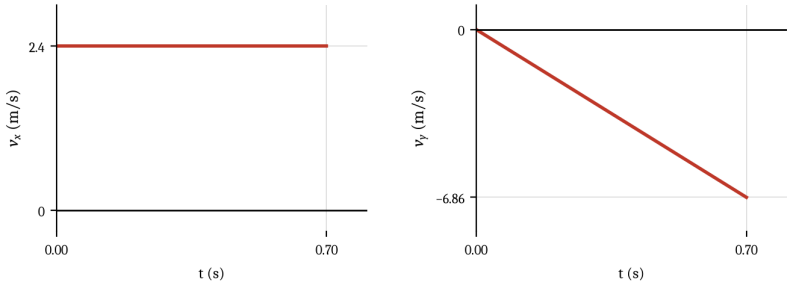
5. ○ Halfway through its time in the air, has a horizontally launched ball fallen half the height? Explain.

6. ● Area under a velocity–time graph. The graphs show a ball that rolls off a ledge and lands on the ground below (up positive).

a) Use the area under the v_x–t graph (a rectangle) to find how far from the ledge the ball lands.

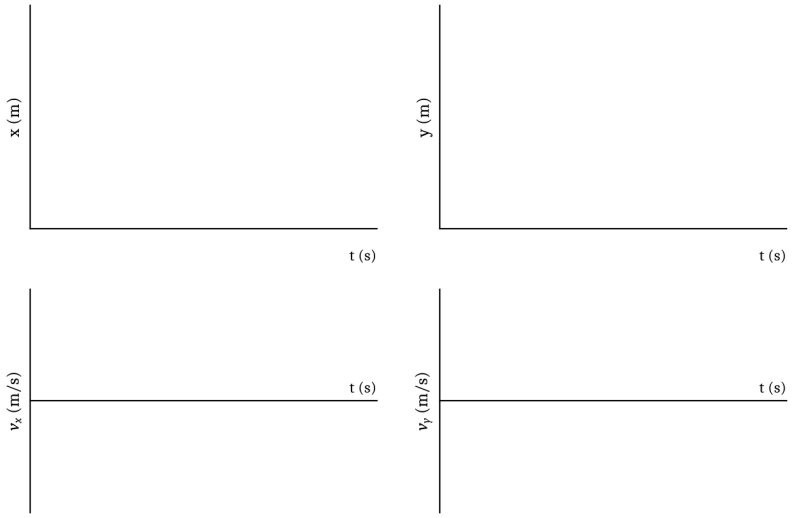
b) Use the area under the v_y–t graph (a triangle) to find Δy. How high is the ledge?

c) Use the slope of the v_y–t graph to find a_y.



7. ● Motion graphs. Sketch all four graphs for a marble that rolls off a 0.85 m table at 1.5 m/s, from the moment it leaves the table until it lands. Put $x_i = 0$ at the table edge and $y = 0$ on the floor, with up positive. Label the starting value on each graph and mark the landing time $t_L (t_L = 0.416\ \text{s})$.

Then: what is the slope of the x–t graph? Of the v_y–t graph?



8. ★ Challenge. A marble leaves a 0.80 m table at 2.0 m/s. Find its velocity (speed and direction) just before it lands.

9. ● Check yourself. During a horizontal launch, which velocity component changes and which stays the same? Why?

Section 2.9 · Marble off the Table Lab

Tier: Core Skills **In class:** Marble off the Table lab

Learning targets

- C2 I can organize a projectile problem into x and y columns linked by the shared time.
- C3 I can predict where a horizontally launched marble will land, then test the prediction.

Key ideas

- Measure the launch speed on the tabletop: $v_{ix} = \dfrac{\text{distance rolled}}{\text{time}}$
- Measure the height from the floor to the tabletop: that is y_i. The floor is $y_f = 0$.
- Predict with the columns, then test once. $\%$ difference $= \dfrac{|\text{predicted} - \text{actual}|}{\text{predicted}} \times 100$

Worked example: the prediction

A marble rolls 0.60 m along the tabletop in 0.40 s, then leaves the 0.85 m table. Where should the cup go? The marble actually lands 0.60 m out. Find the percent difference.

1. Launch speed: $v_{ix} = 0.60 \text{ m} \div 0.40 \text{ s} = 1.5 \text{ m/s}$.
2. y column: $0 = -4.9t^2 + 0.85$. Desmos crossing: $t = 0.416$ s.
3. x column: $x_f = (1.5)(0.416) + 0 = 0.625$ m from the point below the table edge.
4. % difference = $|0.625 - 0.60| \div 0.625 \times 100 = 4.0\%$.

Answer: Place the cup 0.625 m from the table edge; 4.0% difference

1. ● **Partly worked.** A marble rolls 0.50 m in 0.25 s, then leaves a 0.90 m table.

a) $v_{ix} = 0.50 \div 0.25 =$ _____ m/s

b) y column: $0 = -4.9t^2 +$ _____, t = _____ s

c) $x_f =$ _____ × _____ = _____ m

d) It lands 0.82 m out. % difference = _____

2. ● **Your lab data.** Record your group's measurements and prediction.

Quantity	Value
Distance rolled on the table (m)	
Time to roll it (s)	
Launch speed v_{ix} (m/s)	
Table height y_i (m)	
Time in the air t (s)	
Predicted landing x_f (m)	
Actual landing (m)	
% difference	

3. ● Show your prediction work for the table above: both columns, the Desmos crossing, and x_f.

4. ● Name two sources of error in the lab. For each, say whether it would make the marble land closer or farther than predicted.

5. ● A group predicts 0.75 m and the marble lands 0.71 m out. Find the percent difference.

6. ○ Another group's marble leaves the 0.85 m table at 1.2 m/s. Where should they place the cup?

7. ★ **Challenge.** Your teacher moves the cup to 0.90 m from the edge of the 0.85 m table. How fast must the marble roll to land in it?

8. ● **Check yourself.** Which measurement sets the time in the air, and which sets how far the marble lands?

Get Ready for Quiz 2

Tier: Quiz 2: Horizontal Launches **In class:** Quiz 2: Horizontal launches (full period)

About this page

- Quiz 2 covers F4, F5, C2, C3, and C4. Complete this page the night before the quiz.

Key ideas

- Write the x and y columns every time, even when the problem seems easy.
- Rate yourself on each target in the tracker before you start.

1. ● A dart is fired horizontally at 18 m/s from 1.6 m above the floor. Find the time in the air and the horizontal distance it travels.

x (horizontal)	y (vertical)
$v_{ix} =$	$v_{iy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =

2. ● For the dart in problem 1, find v_x and v_y just before it hits the floor.

3. ● True or false, and fix the false ones:

a) A projectile's horizontal velocity decreases as it falls.

b) At the moment a ball leaves the table, its acceleration is zero.

c) A heavier ball rolled off a table at the same speed lands at the same spot.

4. ● A ball rolls off a 0.95 m table and lands 1.1 m from the edge. Find its launch speed.

5. ● Ball A is dropped from a table at the same moment ball B rolls off it at 4.0 m/s. Which lands first? Explain.

6. ● A ball thrown horizontally at 10 m/s from a roof lands 15 m from the building. How tall is the building?

7. ● A ball rolls off a shelf at 2.5 m/s. Find v_x and v_y 0.40 s later.

8. ● Sketch the v_x–t and v_y–t graphs for any horizontal launch. What is the slope of each?

9. ○ If a ball rolls off a table twice as fast, what happens to its time in the air and its landing distance?

10. ○ A package is dropped from a plane flying horizontally at 60 m/s, 320 m above the ground. How far ahead of the target should it be released?

11. ○ Ball A rolls off a 1.0 m table at 2.0 m/s. Ball B rolls off a 2.0 m table at 1.0 m/s. Which is in the air longer? Which lands farther out?

12. ★ **Challenge.** For the dart in problem 1, find the speed and direction just before it hits the floor.

Section 2.10 · Angled Launches: Split and Symmetry

Tier: Target **In class:** Split the launch; symmetry; level-ground practice

Learning targets

- T1 I can find the time in the air and the range of a projectile launched at an angle on level ground.

Key ideas

- Split the launch velocity first: $v_{ix} = v_i \cos\theta$ $v_{iy} = v_i \sin\theta$
- At the top, $v_y = 0$ but v_x is unchanged. The acceleration is still −9.8 m/s².
- Time to the top: $0 = a_y t + v_{iy}$ On level ground, time up = time down, so double it for the total time.
- Range: $x_f = v_{ix} t + x_i$ with the total time.

Worked example

A ball is kicked at 20.0 m/s at 35° above level ground. Find the time in the air and the range.

1. Split: $v_{ix} = 20.0 \cos 35^\circ = 16.4$ m/s; $v_{iy} = 20.0 \sin 35^\circ = 11.5$ m/s.
2. Time to the top: $0 = (-9.8)t + 11.5$, so $t = 1.17$ s.
3. Total time: $2 \times 1.17 = 2.34$ s.
4. Range: $x_f = (16.4)(2.34) + 0 = 38.4$ m.

Answer: In the air 2.34 s; lands 38.4 m away

1. ● **Partly worked.** A ball is launched at 25.0 m/s at 30° above level ground.

a) $v_{ix} = 25.0 \cos 30^\circ =$ _____ m/s $v_{iy} = 25.0 \sin 30^\circ =$ _____ m/s

b) Time to the top: $0 = (-9.8)t +$ _____, t = _____ s

c) Total time = 2 × _____ = _____ s

d) Range: $x_f =$ _____ × _____ + 0 = _____ m

2. ● Split each launch velocity into v_{ix} and v_{iy}.

a) 30.0 m/s at 60° b) 8.0 m/s at 25° c) 50.0 m/s at 10°

3. ● An arrow is shot at 45.0 m/s at 20° above level ground. Find the time in the air and the range.

4. ○ A ball is thrown at 12.0 m/s at 50° above level ground. Find the time in the air and the range.

5. ● Two balls are kicked at 15.0 m/s on level ground, one at 30° and one at 60°. Find both ranges. Which one stays in the air longer?

6. ○ A punter kicks a football at 16.0 m/s at 55° above level ground. Find its time in the air and its range.

7. ● A ball launched from level ground lands 2.4 s later. Its horizontal velocity is 10.0 m/s.
a) How long does it take to reach the top? b) What is v_{iy}? c) What is its range?

8. ● At the highest point of an angled launch, what are v_y, v_x, and the acceleration? Explain.

9. ○ A ball is launched at 10.0 m/s at 45° above level ground. Find its time in the air and its range.

10. ★ **Challenge.** A ball is in the air for 3.0 s and lands 45 m away on level ground. Find v_{ix} and v_{iy}, then its launch speed and angle.

11. ● **Check yourself.** Why can you double the time to the top to get the total time? When would that not work?

Section 2.11 · Level-Ground Launches with Desmos

Tier: Target **In class:** Desmos for time and maximum height; comparing angles

Learning targets

- T1 I can find the time in the air, the maximum height, and the range of a level-ground angled launch, and check them with Desmos.

Key ideas

- Maximum height: at the top $v_{fy} = 0$, so $0 = v_{iy}^2 + 2a_y(y_{max} - 0)$
- Desmos: graph $y = -4.9x^2 + v_{iy}x$ The x-axis crossing is the time in the air; the top of the curve is (time to the top, maximum height).
- Steeper launches go higher and stay up longer. 45° gives the longest range on level ground; angles that add to 90° land in the same spot.

Worked example

A ball is kicked at 15.0 m/s at 40° above level ground. Find the time in the air, the maximum height, and the range.

1. Split: $v_{ix} = 11.5$ m/s; $v_{iy} = 9.64$ m/s.
2. Desmos: graph $y = -4.9x^2 + 9.64x$. It crosses the x-axis at 1.97, so $t = 1.97$ s.
3. Click the top: (0.984, 4.74), so the maximum height is 4.74 m.
4. By hand check: $0 = 9.64^2 + 2(-9.8)(y_{\max} - 0)$, $y_{\max} = 4.74$ m.
5. Range: $x_f = (11.5)(1.97) + 0 = 22.6$ m.

Answer: 1.97 s in the air; 4.74 m high; lands 22.6 m away

1. ● **Partly worked.** A ball is kicked at 20.0 m/s at 35° above level ground.

a) v_{ix} = _____ m/s v_{iy} = _____ m/s

b) Desmos equation: $y = -4.9x^2$ + _____x

c) Crossing: t = _____ s Top: (_____ s, _____ m)

d) Range: x_f = _____ × _____ + 0 = _____ m

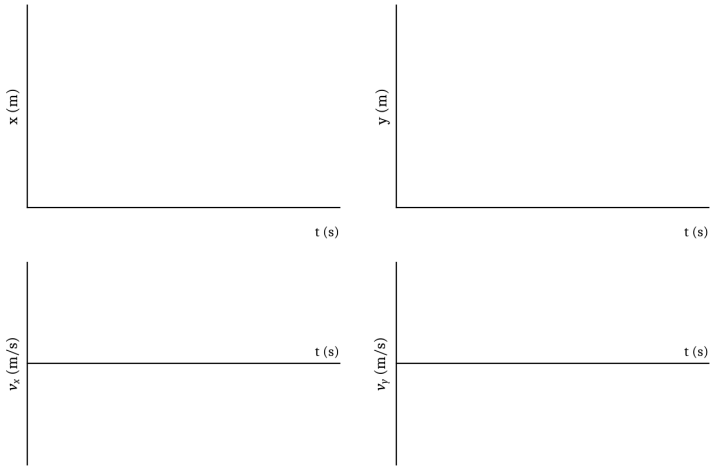
2. ● A football is kicked at 22.0 m/s at 50° above level ground. Use Desmos to find the time in the air and the maximum height. Then find the range.

3. ○ A golf ball leaves the tee at 40.0 m/s at 15° above level ground. Find its time in the air, maximum height, and range.

4. ● Complete the table for a ball launched at 15.0 m/s from level ground. Then answer: which angle gives the longest range? Which two angles give the same range?

Angle	v_{iy} (m/s)	Time in the air (s)	Max height (m)	Range (m)
30°				
45°				
60°				

5. ● **Motion graphs.** Sketch x–t, y–t, v_x–t and v_y–t for the football in problem 2. Mark the time of the top with a dashed line and label the values at launch and landing.



6. ● In Desmos, a student graphs $y = -4.9x^2 + 14.7x$. What is v_{iy}? When does the ball land? How high does it go?

7. ★ **Extension.** A ball is thrown at 15.0 m/s at 30° from the top of a 10.0 m cliff. Graph $y = -4.9x^2 + v_{iy}x + 10$ in Desmos to find the time in the air, then find how far from the base it lands.

8. ● **Check yourself.** On the Desmos graph of an angled launch, what does the x-axis crossing tell you? What does the top of the curve tell you?

Unit 2 Review

Tier: All tiers **In class:** Review stations by tier

About this page

- Use your target tracker to choose where to start. Begin with the lowest tier that is not yet secure.

Key ideas

- Foundations: problems 1–4. Core Skills: problems 5–9. Target: problems 10–11. Everyone: problems 12–14. Extra practice: 15–17.
- Check your work against the key your teacher posts, then redo one missed problem from the matching section.

1. Foundations. Find the resultant of 7.0 km east and 24 km north.

2. Foundations. Resolve 40 m/s at 65° above the horizontal into components.

3. Foundations. A ball is thrown horizontally off a cliff. Describe what happens to its horizontal velocity, vertical velocity, and acceleration during the fall.

4. Foundations. You walk 30 m east and then 30 m west. Find your distance traveled and your change in position.

5. Core Skills. A student walks 6 blocks north, 4 blocks east, 2 blocks south, and 3 blocks west. Find the total change in position (size and direction).

6. Core Skills. A ball rolls off a 2.2 m ledge at 4.0 m/s. Find the time to land and where it lands.

7. Core Skills. For the ball in problem 6, find v_x and v_y at $t = 0.30$ s.

8. Core Skills. A kayak heads straight across a 50 m river at 2.0 m/s relative to the water with a 0.80 m/s current. Find the time, the drift, and the velocity relative to the bank.

9. Core Skills. A marble rolls off a 0.90 m table and lands 1.20 m away. Find its launch speed.

10. Target. A ball is kicked at 30 m/s at 40° on level ground. Find its time in the air, maximum height, and range.

11. Target. A student graphs $y = -4.9x^2 + 19.6x$ in Desmos for a level-ground kick. What is v_{iy}? When does the ball land? How high does it go?

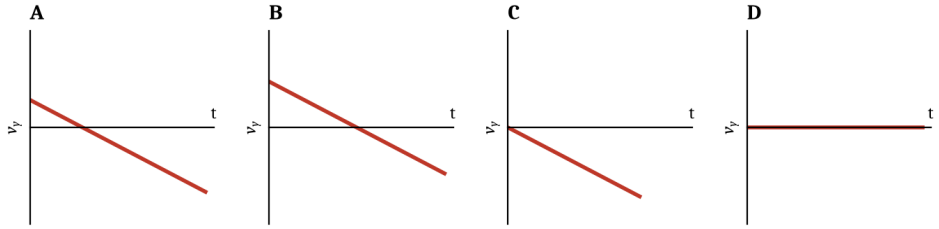
12. Graph matching (everyone). Each graph shows v_y vs. t (up positive) for one motion, from launch to landing. Match each motion to a graph.

1) a marble rolling across a level tabletop 2) a ball rolling off a table

3) a ball kicked from level ground that lands at the same height

4) a ball thrown upward at an angle from a cliff that lands below the cliff

Matches: 1 ___ 2 ___ 3 ___ 4 ___



13. Paragraph argument (everyone). Two balls are launched at 26 m/s from level ground, one at 30° and one at 60°. A classmate claims: "They land at the same spot, so they must be in the air for the same time." Do you agree? Write a paragraph argument: a claim, evidence (a physics principle or equation), and reasoning that links them, in 3–5 complete sentences.

14. Challenge (everyone). Two balls leave a 1.5 m table at the same moment: one rolls off at 2.0 m/s, the other is dropped from rest. How long until they hit the floor, and how far apart are they then?

15. Extra practice (Core Skills). A plane points north at 120 m/s while a 35 m/s wind blows east. Find the plane's velocity relative to the ground.

16. Extra practice (Target). A soccer ball is kicked at 18 m/s at 55° on level ground. Find its time in the air and its range.

17. Challenge. Two forces of 5 N and 12 N can point any way. What are the largest and smallest resultants possible? At what angle between them is the resultant 13 N?

Projectile Launch Project: Planning Pages

Tier: Target **In class:** Projectile Launch Project, Parts 1 to 3

About this page

- Use these pages with the Projectile Launch Project handout. Part 1 finds the launch speed, Part 2 predicts the landing spot, and Part 3 is the one-shot launch.

Key ideas

- Part 1: fire the launcher horizontally off a table. The y column gives the time from the table height; then $v_i = \dfrac{x_f}{t}$ from the average landing distance. Only a meter stick is allowed.
- Part 2: split v_i at the given angle. The launcher sits on the floor, so the ball lands at the height it started. Graph $y = -4.9x^2 + v_{iy}x$ in Desmos. The crossing is the time in the air, the top is the maximum height, and $x_f = v_{ix}t + x_i$ is where the target goes.
- Part 3: one launch. Compare with $\%\ \text{difference} = \dfrac{|\text{predicted} - \text{actual}|}{\text{predicted}} \times 100$

1. Part 1 rehearsal. A launcher fires horizontally off a 1.10 m table. Five shots land 2.31, 2.27, 2.35, 2.29, 2.33 m from the point below the launcher.

a) Find the average landing distance. b) Find the time in the air. c) Find the launch speed.

2. Part 1: your data. Fill in the table on project day 1.

Quantity	Value
Table height y_i (m)	
Landing distance, trial 1 (m)	
Trial 2 (m)	
Trial 3 (m)	
Trial 4 (m)	
Trial 5 (m)	
Average landing distance (m)	
Time in the air t (s)	
Launch speed v_i (m/s)	

3. Part 2 rehearsal. The same launcher (4.88 m/s) sits on the floor and fires at 35° above the horizontal. The ball lands on the floor.

a) Find v_{ix} and v_{iy}. b) Write the Desmos equation for y. c) Read the time in the air and the top of the graph. d) Where should the target go?

4. Part 2: your prediction. Launch angle $\theta =$ _____° v_i (from Part 1) = _____ m/s

$v_{ix} =$ _____ m/s $v_{iy} =$ _____ m/s Desmos equation: y = _____

Time in the air t = _____ s Maximum height $y_{\max} =$ _____ m Target distance $x_f =$ _____ m

5. Sources of error. Name two measurements that could throw off your target placement and how you will reduce each one.

6. Part 3: results. Predicted x_f: _____ Actual x_f: _____ Percent difference: _____

What most likely explains the difference?

7. Percent difference practice. A group predicted a landing at 2.28 m and measured 2.19 m. Find the percent difference.

8. Think it through. A group is assigned 55° instead of 35°, with the same launcher. Will the ball land closer, farther, or in the same spot? Explain in one sentence.

9. Extension bonus. Your teacher moves the launcher onto a 1.10 m table, still at 35° and 4.88 m/s. Use Desmos (add + 1.10 to your equation) to find where the target should go now.

Target tracker

After each section and before each quiz, rate yourself: ✓ secure, ~ getting there, × need help. Write the quiz or reassessment result when you get it back.

Tier	Target	I can...	Section	Self-rating	Quiz / test evidence
Foundations	F1	Draw vectors to scale and add them tip-to-tail	2.2		
Foundations	F2	Resolve a vector into x and y components (SOH-CAH-TOA)	2.3		
Foundations	F3	Find a resultant's magnitude and direction	2.2–2.4		
Foundations	F4	Explain why horizontal and vertical motions are independent	2.6		
Foundations	F5	State v_x is constant and $a_y = -9.8\ \text{m/s}^2$	2.6		
Foundations	F6	Find the resultant of two perpendicular vectors	2.2		
Core Skills	C1	Add a trip of north, south, east, and west legs using x and y parts	2.4		
Core Skills	C2	Organize a projectile problem into x and y columns	2.6		
Core Skills	C3	Solve horizontal launches: time to land, landing position	2.6–2.7		
Core Skills	C4	Find velocity components at any time	2.8		
Core Skills	C5	Relative velocity with perpendicular vectors	2.5		
Target	T1	Angled launch on level ground: time, max height, range	2.10–2.11		
Target	C1+	Add vectors at any angle using components	2.4 Challenge		

Extension menu

Open to students who have shown the Target tier. Choose one. The Projectile Launch Project is the class-wide predict-then-test lab; these go further.

Aiming upstream

Find the angle a boat must aim upstream to land directly across a river, and how long the crossing takes. Then do the same for a plane flying in a crosswind.

Launches from a height

Solve angled launches from a cliff or onto a ledge with Desmos, and find the velocity (speed and angle) just before landing.

Air resistance

Film a badminton birdie or foam ball, track its path with video analysis, and compare it to the ideal model. Explain where and why the real path differs.

My choice: _____ Due date: _____